Student Name	**Janet Oluwatosin Michael**
Lab Title	**Proactive-Threat-Hunting-Security-Incident-Investigation-and-Active-Response-Using-Wazuh**
Course Code	**CIP-A107**
Registration Number	**2026/EHIT/17073**
Cohort / Programme	**Cohort 11**
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Instructor / Assessor	**Mr Aminu Idris**

1. Introduction

The purpose of this project was to practise the work of a Level 2 Security Operations Centre (SOC) analyst using Wazuh. The lab involved investigating suspicious login activities, checking process execution, reviewing PowerShell activity, identifying possible indicators of compromise, and configuring Active Response to block a suspicious IP address.

I started the project on 25 October 2026. My lab environment consisted of an Ubuntu machine running the Wazuh manager and a Windows 11 virtual machine connected as the Wazuh agent. The Windows agent had ID 001 and IP address 192.168.0.247, while the Ubuntu manager was named janet-VirtualBox and used IP address 192.168.0.2.

I carried out the investigation by searching the Wazuh Security Events, reviewing Windows security events, testing PowerShell logging, and generating controlled SSH login attempts. I also checked the manager logs and firewall rules to confirm the result of the Active Response test.

1.1 Lab Environment

The tools and systems I used for this project were:

- Wazuh Manager installed on Ubuntu.
- Wazuh Dashboard for viewing security alerts.
- Windows 11 virtual machine as the Wazuh agent.
- SSH for testing authentication attempts against the Ubuntu server.
- Linux authentication logs for checking login activity.
- Firewall rules for verifying the active response.

The Ubuntu server had the IP address 192.168.0.2, while the Windows 11 machine had the IP address 192.168.0.247. The Windows agent was registered in Wazuh with agent ID 001.

2. Part 1: Threat Hunting and Log Analysis

2.1 Investigation of failed logon events

I started by searching the Wazuh Security Events for Windows failed logon events using Event ID 4625. This event records an unsuccessful attempt to log in to a Windows system.

One of the events I found had the following details:

Field	**Observed value**
Event ID	4625
Wazuh rule ID	60122
Rule level	5
Windows event time	25 October 2026, 15:17:21 UTC
Wazuh alert time	25 October 2026, 16:17:35 UTC+1
Logon type	2 — Interactive
Source address	192.168.0.247
Status	0xc000006d
Substatus	0xc0000380
Process	svchost.exe

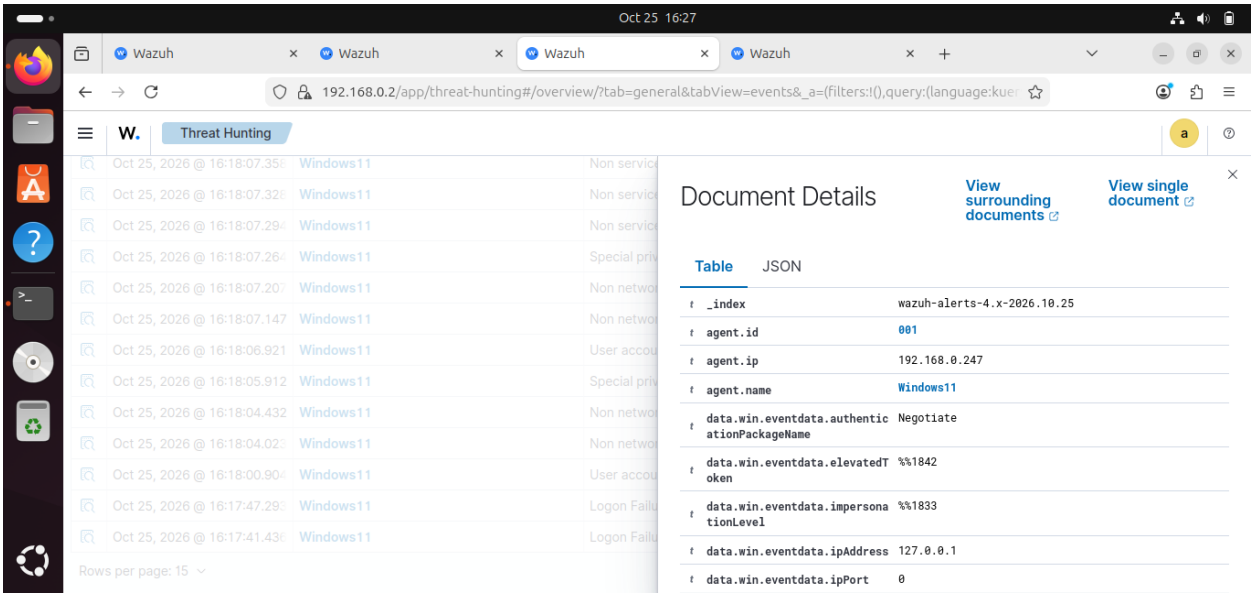
The event showed that a logon attempt had failed.

2.2 Investigation of successful logon events

I then searched for Event ID 4624, which records a successful logon.

The successful logon associated with the earlier investigation had these details:

Field	Observed value
Event ID	4624
Wazuh rule ID	67022
Windows event time	25 October 2026, 15:17:46 UTC
Wazuh alert time	25 October 2026, 16:18:04 UTC+1
Account	janet.michael16@outlook.com
Logon type	11 — Cached Interactive
Source address	192.168.0.247



The successful logon was recorded approximately 25 seconds after the failed logon event. The timing made it worth checking whether the events were related.

I also observed another successful logon event with rule ID 60106, level 3, running under SYSTEM with logon type 5 and process services.exe. That event was consistent with a service logon and was not confirmed to be malicious.

This part of the investigation taught me that a successful logon following a failed logon should not automatically be treated as a compromised account. The analyst needs to compare the account, source address, logon type, timestamps, and other activity before deciding whether the events are connected.

2.3 Process execution investigation

After reviewing the logon events, I searched for Windows process creation events using Event ID 4688. This event is useful because it records the creation of a new process.

I confirmed that Wazuh was receiving process creation events from the Windows machine. During the investigation, I reviewed events involving cmd.exe and conhost.exe. One event showed cmd.exe with svchost.exe as its parent process, but the command line was blank. I also observed a conhost.exe process associated with PowerShell activity.

These events helped me understand how process creation can provide more information about what happens on an endpoint. However, the process details available in my evidence did not prove that the processes were malicious or that they were directly connected to the earlier logon events.

3. Part 2: PowerShell Investigation and IOC Hunting

3.1 PowerShell Script Block Logging

The next part of the lab required me to investigate PowerShell activity. I looked at Event ID 4104, which records PowerShell script blocks when the appropriate logging is enabled.

I tested the PowerShell logging process on my Windows 11 machine and checked Windows Event Viewer. The test event appeared locally in Windows Event Viewer, but I did not confirm that the same Event ID 4104 test was received by Wazuh.

This was an important finding because it showed that seeing an event on the Windows endpoint does not necessarily mean it has been successfully collected and displayed in the Wazuh dashboard. The event collection and configuration must also be checked.

The lab scenario described a Base64-encoded command associated with PowerShell activity. However, I do not have the exact encoded command in the evidence available for this report, so I cannot provide a reliable decoded result.

3.2 Simulated file creation and File Integrity Monitoring

The lab instructions also described a simulated attack involving the file payload.exe in the SensitiveFiles directory. The purpose was to test whether Wazuh File Integrity Monitoring could detect a new file in a monitored location.

I worked on the SensitiveFiles directory as part of the lab. However, the evidence available to me does not confirm a corresponding Wazuh File Integrity Monitoring alert for payload.exe. I therefore cannot state that Wazuh detected the file creation as a confirmed result.

This part showed me why File Integrity Monitoring is important in a SOC environment. If an unexpected file appears in a sensitive directory, an analyst can investigate the filename, creation time, file hash, and related process activity. These details can help determine whether the file is legitimate or suspicious.

3.3 Correlation of the Windows events

The intended scenario described a sequence of failed logons, a successful logon, PowerShell execution, and an unauthorised file creation. I investigated the Windows logon and process events, and I tested PowerShell logging.

However, I could not confirm every stage of that sequence from the collected Wazuh evidence. In particular, the logon events used a loopback source address, the relationship between the logons and process events was not established, and the PowerShell and file creation tests did not produce confirmed Wazuh alerts in the evidence reviewed.

For that reason, I treated the Windows investigation as a series of events that required analysis rather than claiming that the complete attack sequence had been proven.

4. Part 3: Active Response and Threat Containment

4.1 Configuring Active Response

For Part 3, I configured Wazuh Active Response on the Ubuntu manager. The purpose was to automatically block a source IP address after repeated suspicious SSH login attempts.

The assignment provided a configuration using firewall-drop.sh, location manager, and rule ID 5715. When I tested the configuration, my installed Wazuh version rejected manager as an active response location. I changed the location to server, and the configuration test completed without displaying an error.

I also checked the installed Wazuh ruleset and found that rule 5715 represented successful SSH authentication on my installation. It was not the rule for the repeated failed login activity I was testing. I therefore configured the response to use rule 5712, which detects repeated attempts involving a nonexistent username.

The response was configured with a timeout of 600 seconds, which is 10 minutes. My installed script was named firewall-drop, without the .sh extension, so I used the executable available on the system.

After making the changes, I reloaded the systemd service definitions, restarted Wazuh Manager, and confirmed that the manager was active. I also ran the Wazuh analysis configuration test, which returned no error.

4.2 Generating the SSH test activity

To test Active Response, I used the Windows 11 machine to attempt an SSH connection to the Ubuntu manager. The command used was:

ssh fakeuser123@192.168.0.2

The username fakeuser123 was a test username that did not exist on the Ubuntu system.

The Ubuntu authentication logs recorded invalid-user and failed-password events from 192.168.0.247. Wazuh detected repeated attempts involving this username. The relevant detection rule was 5712, which had a level of 10 and required eight matching events within 120 seconds from the same source IP address.

4.3 Active Response execution evidence

At 22:13:43 on 25 October 2026, the Active Response log recorded the following activity:

- The firewall-drop script started.
- Wazuh issued an add command for the alert associated with rule 5712.
- The alert identified source IP address 192.168.0.247.
- The response script processed the IP address through its check_keys operation.
- The script ended.

The alert description was:

sshd: brute force trying to get access to the system. Non existent user.

This was evidence that Wazuh detected the repeated SSH activity and triggered the configured Active Response script.

4.4 Firewall verification

After the script ran, I executed:

sudo iptables -L -n

The output showed a DROP rule for 192.168.0.247 in the INPUT chain and another DROP rule for the same address in the FORWARD chain.

This confirmed that firewall rules to drop traffic from the Windows lab machine's IP address were present when I checked the firewall. The response had been configured for a 600-second timeout, although I did not separately verify the exact time when the rules were removed.

The Active Response test was the clearest confirmed containment result in my project because I had both the Wazuh response log and the firewall output as evidence.

5. Indicators of Compromise

The following table lists the indicators relevant to my investigation and identifies what was confirmed.

Indicator	Value	Finding
SSH source IP	192.168.0.247	Confirmed in the SSH logs and firewall rules
SSH destination	192.168.0.2	Ubuntu Wazuh manager used for the SSH test
SSH username	fakeuser123	Confirmed test username in the SSH authentication logs
Wazuh detection rule	5712	Confirmed in the Active Response log
Windows loopback address	127.0.0.1	Present in the Windows logon event details; not an external attacker address
Payload filename	payload.exe	Specified in the lab scenario; creation was not confirmed in the available evidence
Base64-encoded command	Exact value not available	Not confirmed in the collected evidence

The IP address 192.168.0.247 was the source of the SSH test traffic from my own Windows lab machine. It should therefore be described as the test source IP, rather than an independently identified external attacker.

6. Recommendations

Based on the investigation, I recommend the following actions:

1. Continue monitoring Windows failed and successful logon events, and compare the source address, account, logon type, and timestamps when investigating suspicious activity.
2. Ensure that Windows PowerShell Script Block Logging is enabled and that Event ID 4104 is being collected by the Wazuh agent.
3. Review the File Integrity Monitoring configuration and test whether file creation in SensitiveFiles produces an alert.
4. Investigate any confirmed suspicious executable using its hash, file path, creation time, and related process events.
5. Monitor other endpoints for repeated failed logons, unusual successful logons, and suspicious process execution.
6. Review firewall rules after an Active Response event to confirm that temporary blocks are removed as configured.
7. Reset a user's password and provide security awareness training if an account compromise is confirmed. These actions should not be based solely on the test events in this lab.

Lab Challenge & Advanced Analysis Questions

1. **The Big Picture:** Why is the sequence of "failed logons -> successful logon -> immediate PowerShell execution" so highly suspicious from a behavioral perspective?

Answer : This sequence can be suspicious because an attacker may try different passwords until one works. After gaining access, the attacker may use PowerShell to run commands, download a file, or perform other activities on the system. However, the analyst must compare the timestamps, account details, source IP address, and process information to determine whether the events are related. In my investigation, the Windows evidence did not confirm that all the events belonged to one attack.

2. **Beyond the Alert:** How did proactive threat hunting, as opposed to simply reviewing the initial alert, provide a more complete picture of the incident?

Answer: Proactive threat hunting allowed me to search for events beyond the initial alert. I investigated Event IDs 4625, 4624, and 4688, and I also tested PowerShell logging with Event ID 4104. This helped me understand the login activity and process events available in the lab. I also identified gaps where PowerShell and file creation activity were not confirmed in Wazuh. During Active Response testing, I found that repeated SSH attempts could trigger rule 5712 and cause the firewall-drop script to run. This showed me that investigating more than one type of log helps an analyst understand both the activity and the limits of the available evidence.

3. **Active Response Trade-offs:** What is a potential risk of automatically blocking an IP address for 10 minutes? Can you think of a scenario where this might cause a problem?

Answer : One risk is that the IP address may be shared by several legitimate users. If the address is blocked, those users may also lose access to the service. A temporary block may also interrupt an administrator or other authorised user. This is why an analyst should review the alert and consider the possible effect before automatically blocking an IP address.

4. **The Attacker's Next Move:** If the attacker's IP is now blocked, what is one other technique they might use to regain access to the network, and how could Wazuh help you detect it? (Hint: Think about other endpoints).

Answer: An attacker might try to access another endpoint using stolen credentials or a different source IP address. Wazuh can help detect this by monitoring login events and suspicious process activity across multiple endpoints. If the same account or similar activity appears on another machine, the analyst can investigate whether the activity is connected.

8. Lessons Learned

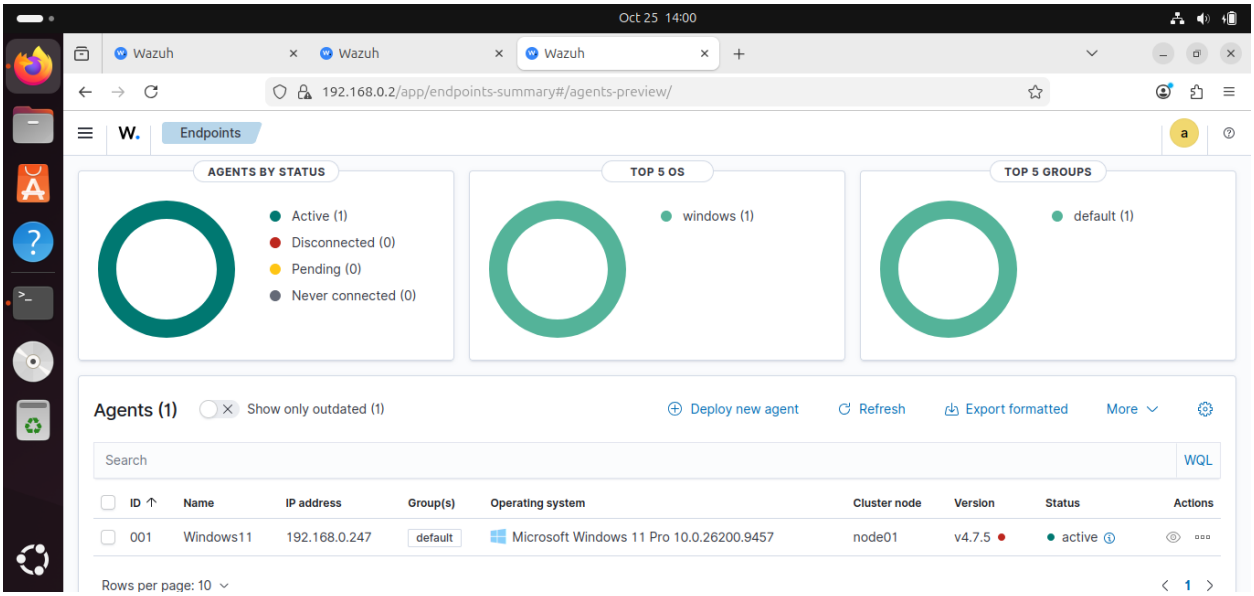
From this project, I learned how to search Windows security events in Wazuh and investigate failed logons, successful logons, and process creation. I also learned that an event should not be labelled malicious without enough evidence to support the conclusion. I tested PowerShell logging and learned that an event appearing in Windows Event Viewer does not automatically mean it has been received by Wazuh.

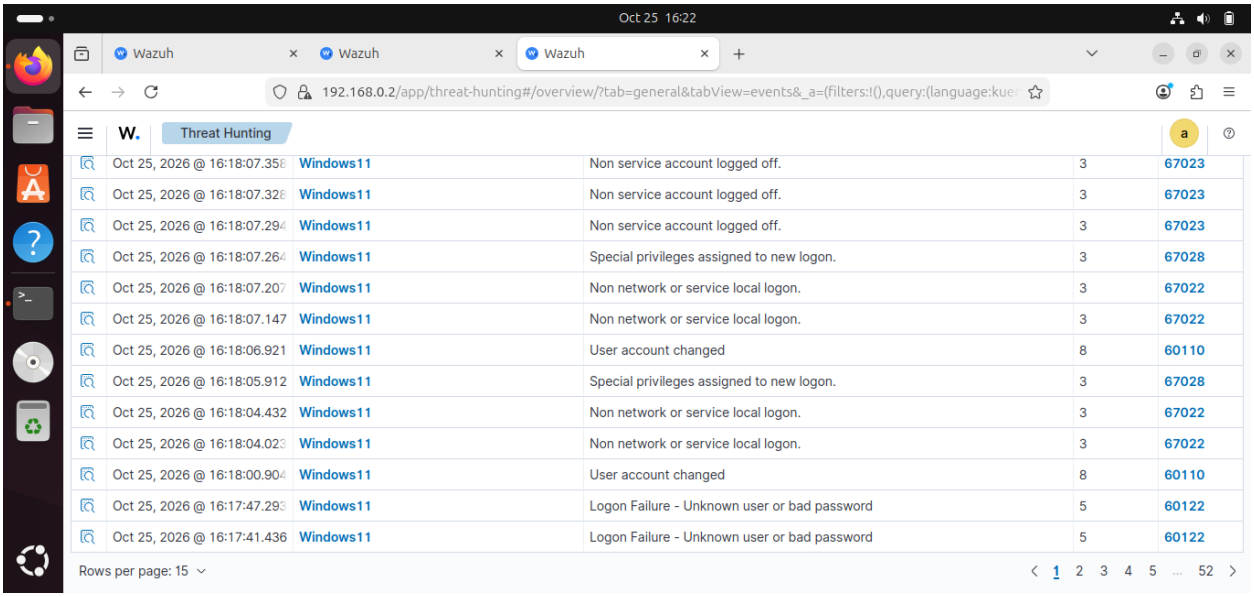
The Active Response part gave me practical experience configuring a response, generating controlled SSH login attempts, checking the response log, and verifying firewall DROP rules. I also learned that a response should be verified instead of assuming that the configuration alone proves that a threat has been blocked. Overall, the project helped me practise the process of collecting evidence, investigating events, taking a response action, and documenting the findings.

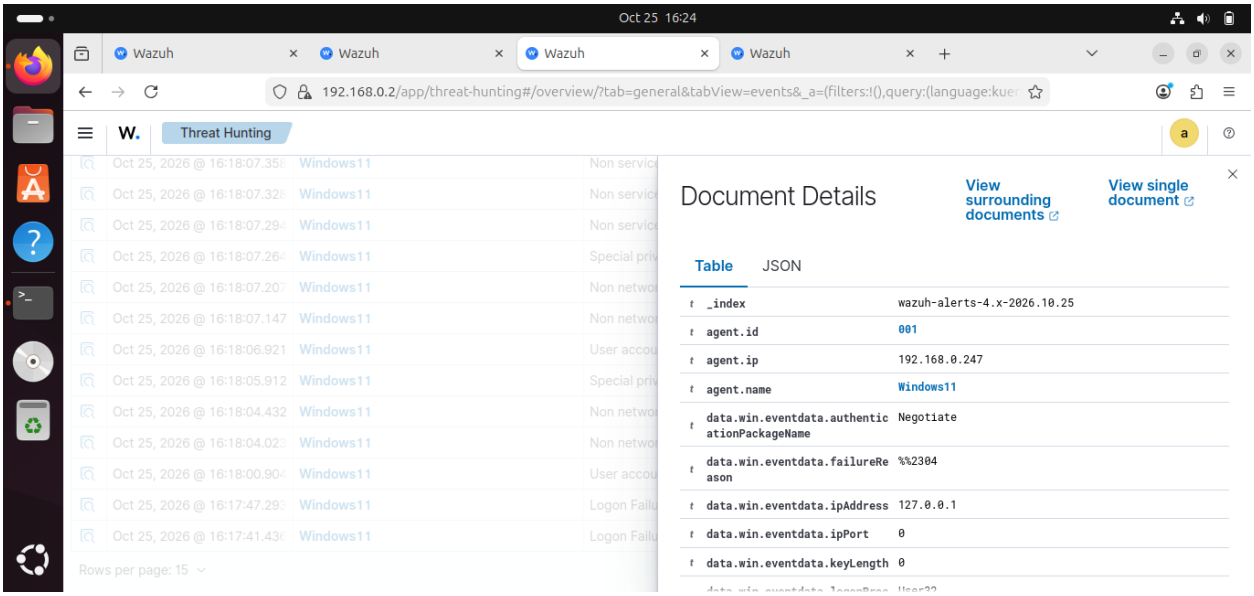
9. Conclusion

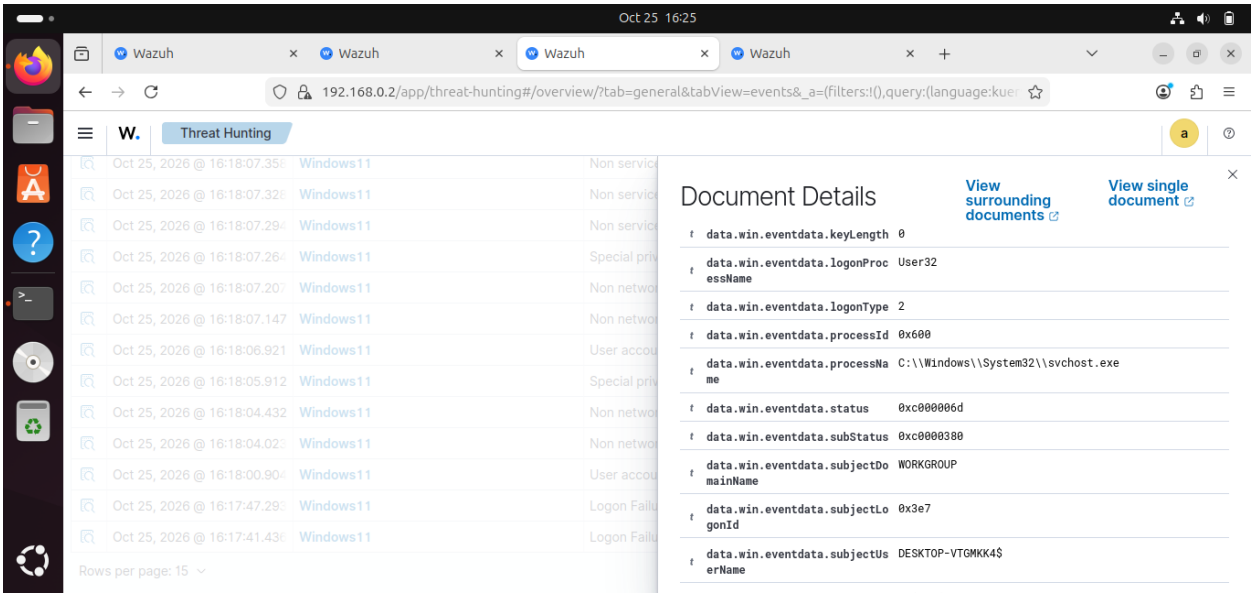
This project allowed me to practise several tasks carried out by a SOC analyst, including threat hunting, Windows event analysis, process investigation, PowerShell logging tests, and Active Response. The Windows investigation helped me understand how to review authentication and process events, while the SSH test confirmed that Wazuh could detect repeated attempts

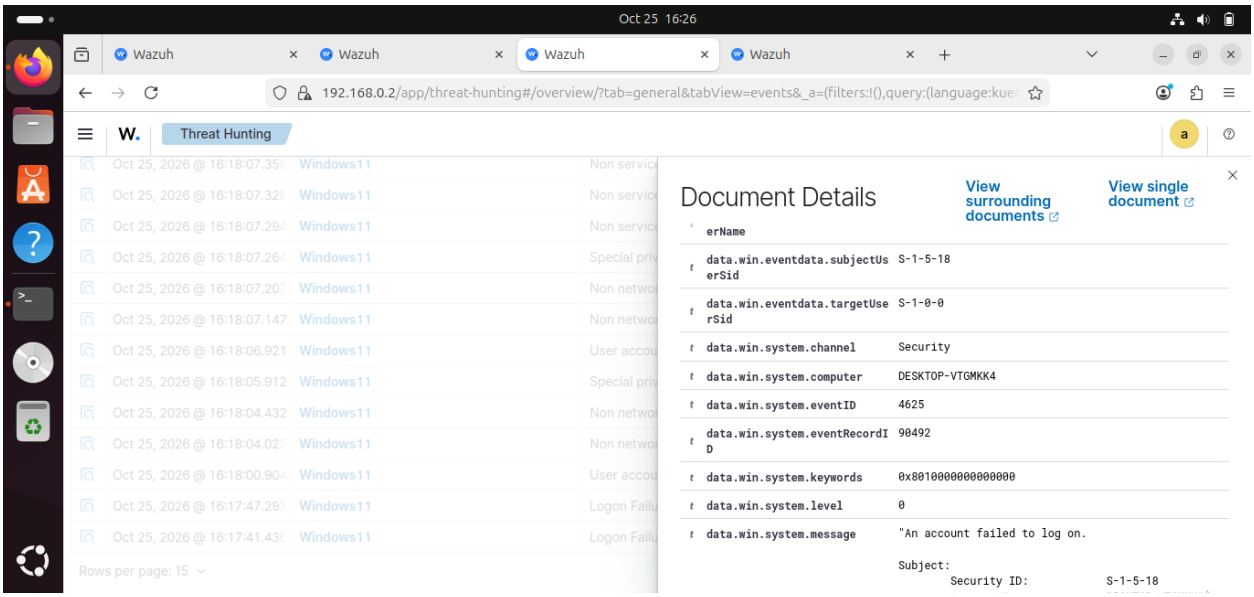
involving a nonexistent username and trigger a firewall response. The firewall output showed DROP rules for the source IP address at the time of verification. Some parts of the original scenario, including the exact Base64 command and confirmed creation of payload.exe, were not established by the evidence available for this report. Recording these limitations is important because an incident report should explain what the evidence proves and what still needs to be investigated.

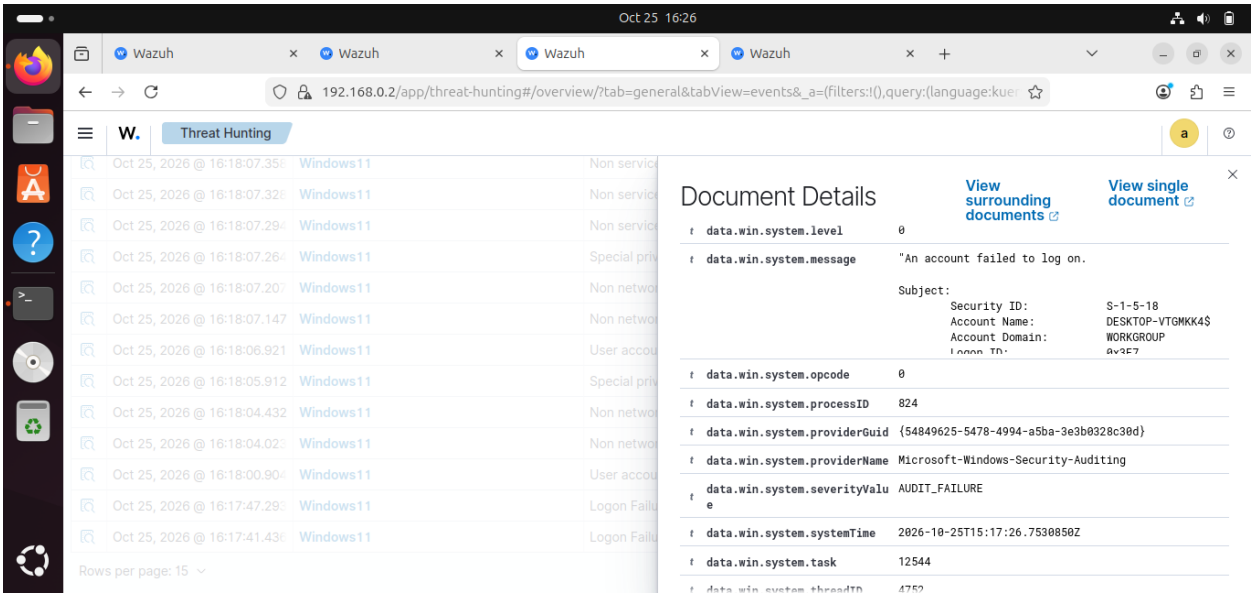


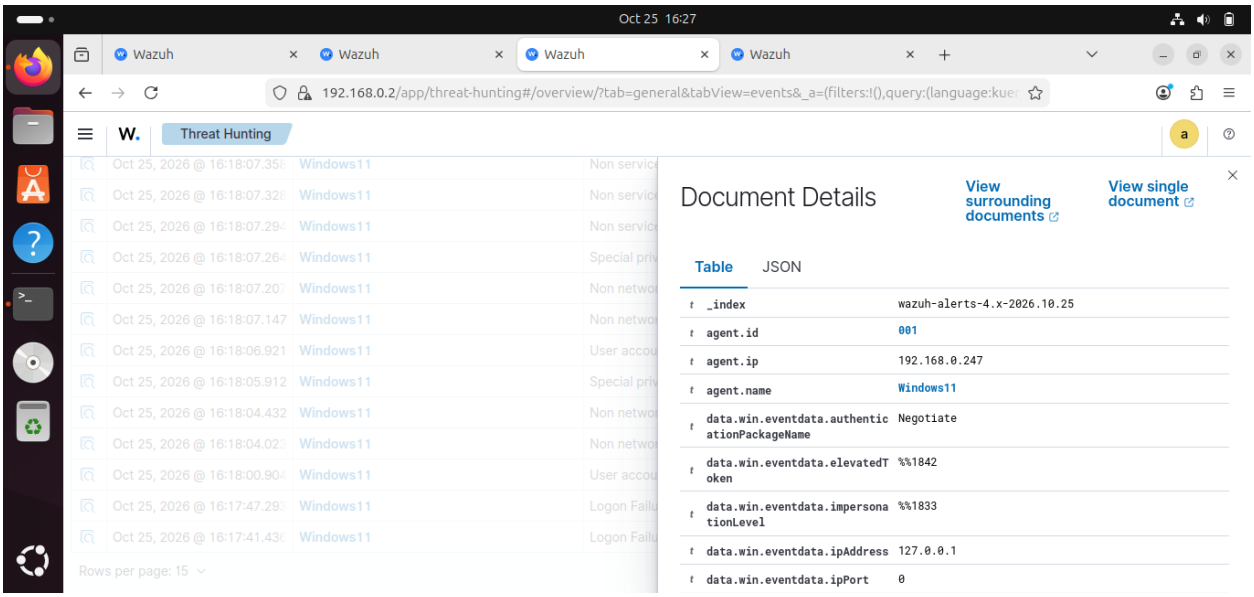

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Oct 25, 2026 @ 16:18:07.294 Windows11 Non service account logged off. 3 67023
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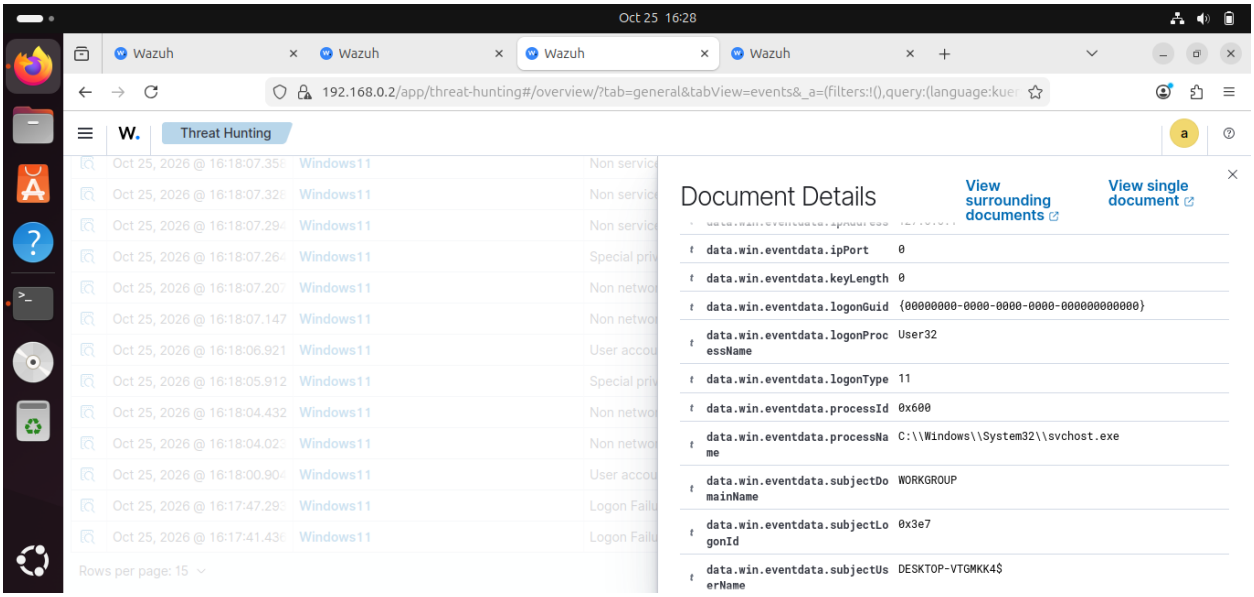

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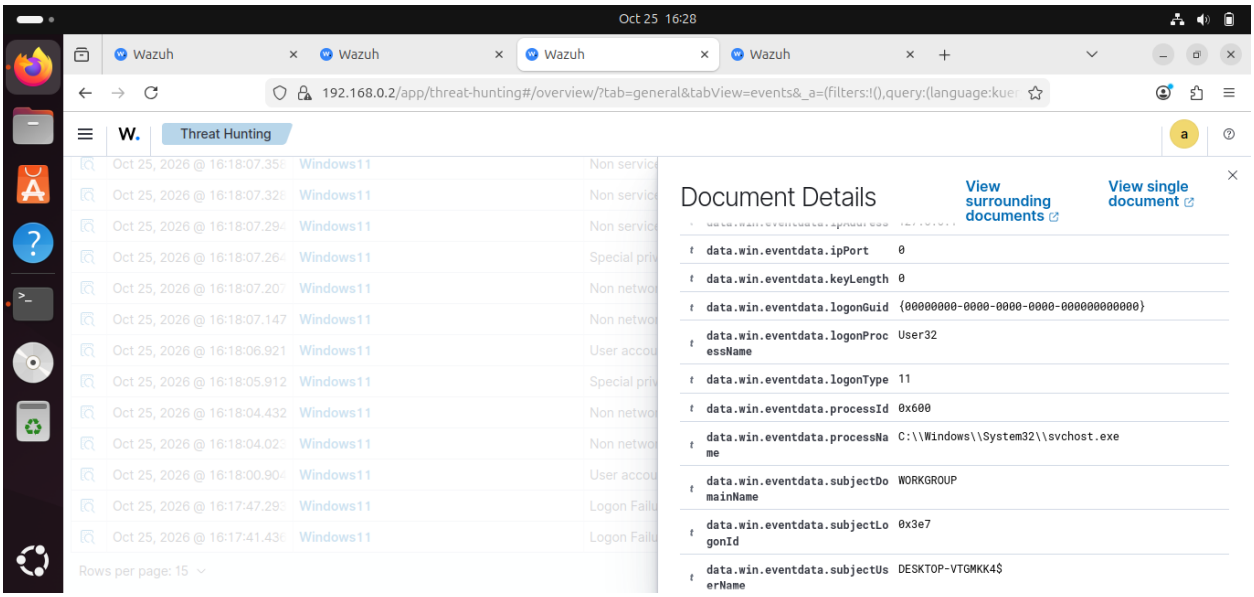

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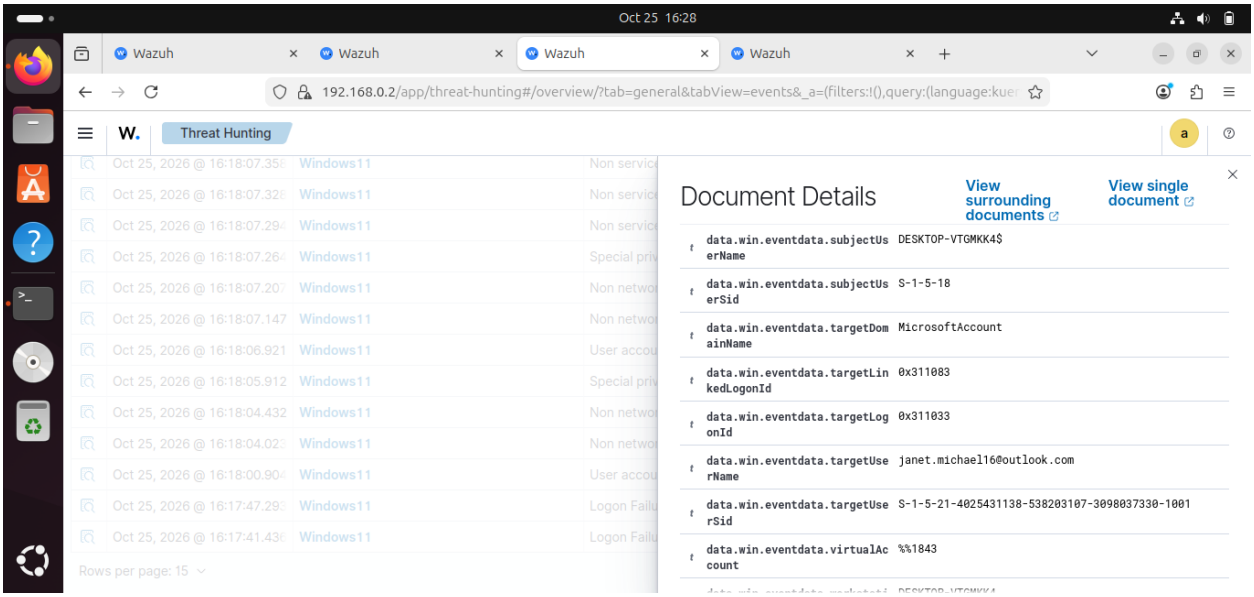

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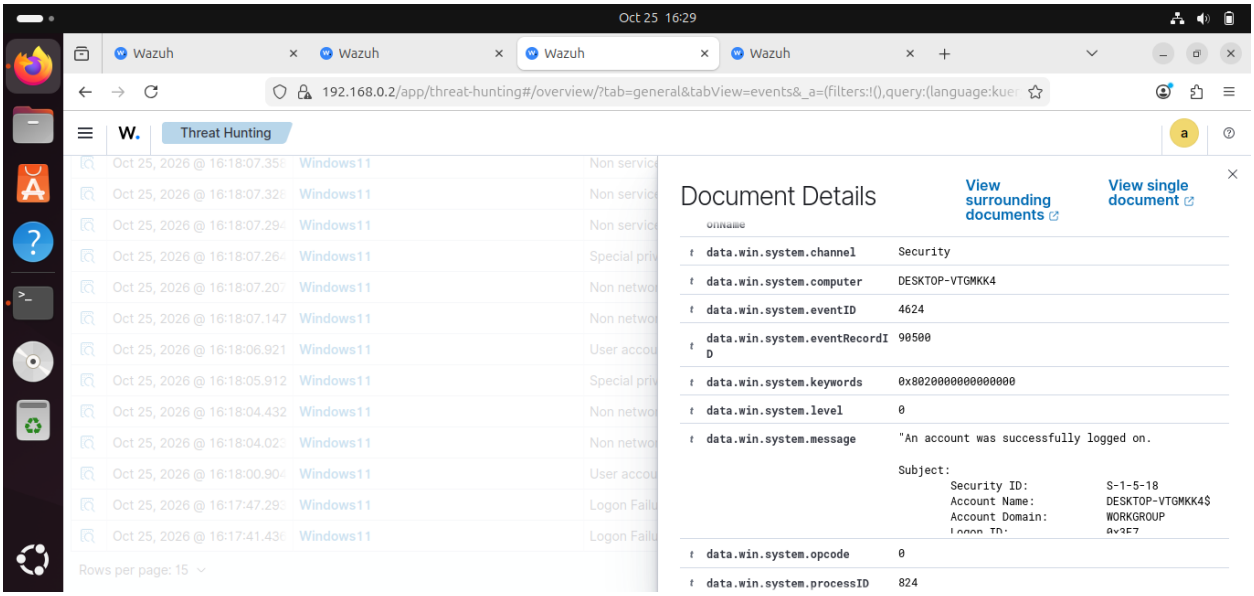

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Account Domain: WORKGROUP
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